Frullani's Theorem

Senan Sekhon

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0 Introduction

Theorem 0.1 (Frullani's Theorem). Suppose $f \in C[0,\infty)$, $\lim_{x\to\infty} f(x) = L$ and $a, b > 0$. Then:

$$\int_0^\infty \frac{f(bx)-f(ax)}{x}\,dx = (L - f(0))\log\left(\frac{b}{a}\right)$$

Proof. If $a = b$, the result holds trivially as both sides are zero. Assume without loss of generality that $a < b$ (otherwise, switch a and b). Suppose $u, v > 0$ and $av > bu$. Then we have:

$$\begin{aligned}
\int_u^v \frac{f(bx)-f(ax)}{x}\,dx &= \int_u^v \frac{f(bx)}{x}\,dx - \int_u^v \frac{f(ax)}{x}\,dx \\
&= \int_{bu}^{bv} \frac{f(x)}{x}\,dx - \int_{au}^{av} \frac{f(x)}{x}\,dx \\
&= \int_{av}^{bv} \frac{f(x)}{x}\,dx + \int_{bu}^{av} \frac{f(x)}{x}\,dx - \int_{bu}^{av} \frac{f(x)}{x}\,dx - \int_{au}^{bu} \frac{f(x)}{x}\,dx \\
&= \underbrace{\int_{av}^{bv} \frac{f(x)}{x}\,dx}_{G(v)} - \underbrace{\int_{au}^{bu} \frac{f(x)}{x}\,dx}_{G(u)}
\end{aligned}$$

We now show that $\lim_{u\to 0^+} G(u) = f(0)\log\left(\frac{b}{a}\right)$. Suppose $\varepsilon > 0$. Then there exists $\delta > 0$ such that $|f(u) - f(0)| < \frac{\varepsilon}{\log(b/a)}$ for all $0 \le u < \frac{\delta}{b}$ (and so $au, bu < \delta$). This yields:

$$\begin{aligned}
\left|G(u) - f(0)\log\left(\frac{b}{a}\right)\right| &= \left|\int_{au}^{bu} \frac{f(x)}{x}\,dx - f(0)\log\left(\frac{b}{a}\right)\right| = \left|\int_{au}^{bu} \frac{f(x)}{x}\,dx - f(0)\int_{au}^{bu} \frac{1}{x}\,dx\right| \\
&= \left|\int_{au}^{bu} \frac{f(x)-f(0)}{x}\,dx\right| \le \int_{au}^{bu} \frac{|f(x)-f(0)|}{x}\,dx \\
&\le \frac{\varepsilon}{\log(b/a)}\int_{au}^{bu} \frac{1}{x}\,dx = \frac{\varepsilon}{\log(b/a)}\log\left(\frac{b}{a}\right) = \varepsilon
\end{aligned}$$

Thus $\lim_{u\to 0^+} G(u) = f(0)\log\left(\frac{b}{a}\right)$.

We now show that $\lim_{v\to\infty} G(v) = L\log\left(\frac{b}{a}\right)$. Suppose $\varepsilon > 0$. Then there exists $H > 0$ such that $|f(x) - L| < \frac{\varepsilon}{\log(b/a)}$ for all $x > H$. This yields:

$$\begin{aligned}
\left|G(v) - L\log\left(\frac{b}{a}\right)\right| &= \left|\int_{av}^{bv} \frac{f(x)}{x}\,dx - L\log\left(\frac{b}{a}\right)\right| = \left|\int_{av}^{bv} \frac{f(x)}{x}\,dx - L\int_{av}^{bv} \frac{1}{x}\,dx\right| \\
&= \left|\int_{av}^{bv} \frac{f(x)-L}{x}\,dx\right| \le \int_{av}^{bv} \frac{|f(x)-L|}{x}\,dx \le \frac{\varepsilon}{\log(b/a)}\int_{av}^{bv} \frac{1}{x}\,dx \le \frac{\varepsilon}{\log(b/a)}\log\left(\frac{b}{a}\right) = \varepsilon
\end{aligned}$$

Thus $\lim_{v\to\infty} G(v) = L\log\left(\frac{b}{a}\right)$. Finally, we have:

$$\int_0^\infty \frac{f(bx)-f(ax)}{x}\,dx = \lim_{v\to\infty} G(v) - \lim_{u\to 0^+} G(u) = (L - f(0))\log\left(\frac{b}{a}\right)$$

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A simpler proof, assuming f is differentiable and f' is integrable on $[0,\infty)$.

$$\int_0^\infty \frac{f(bx)-f(ax)}{x}\,dx = \int_0^\infty \int_a^b f'(xy)\,dx\,dy = \int_a^b \int_0^\infty f'(xy)\,dy\,dx = \int_a^b \frac{L-f(0)}{y}\,dy = (L - f(0))\log\left(\frac{b}{a}\right)$$

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Frullani's theorem was first stated[1] in 1821 in a letter from Giuliano Frullani (1795–1834) to Giovanni Plana (1781–

[1] http://www.ams.org/journals/proc/1990-109-01/S0002-9939-1990-1007485-4/S0002-9939-1990-1007485-4.pdf

1864), though a complete proof was only given in 1823 by Augustin-Louis Cauchy (1789–1857).

Example. $f(x) = e^{-x}$

$$\int_0^\infty \frac{e^{-bx} - e^{-ax}}{x}\,dx = -\log\left(\frac{b}{a}\right)$$

Example. $f(x) = \arctan(x)$

$$\int_0^\infty \frac{\arctan(bx) - \arctan(ax)}{x}\,dx = \frac{\pi}{2}\log\left(\frac{b}{a}\right)$$

Example. $f(x) = \operatorname{arcsec}(x+1)$

$$\int_0^\infty \frac{\operatorname{arcsec}(bx+1) - \operatorname{arcsec}(ax+1)}{x}\,dx = \frac{\pi}{2}\log\left(\frac{b}{a}\right)$$

Example. $f(x) = \tanh(x)$

$$\int_0^\infty \frac{\tanh(bx) - \tanh(ax)}{x}\,dx = \log\left(\frac{b}{a}\right)$$

For certain values of a and b, this can be rewritten in other forms, e.g.

$$\begin{aligned}\tanh(2x) - \tanh(x) &= \tanh(x)\operatorname{sech}(2x)\\ \tanh(3x) - \tanh(x) &= \frac{2\tanh(x)\operatorname{sech}(x)^2}{1 + 3\tanh(x)^2}\\ \tanh(3x) - \tanh(2x) &= \frac{\tanh(x)\operatorname{sech}(x)^4}{\left(1+\tanh(x)^2\right)\left(1+3\tanh(x)^2\right)}\end{aligned}$$

Example. $f(x) = \operatorname{sech}(x)$

$$\int_0^\infty \frac{\operatorname{sech}(bx) - \operatorname{sech}(ax)}{x}\,dx = -\log\left(\frac{b}{a}\right)$$

For certain values of a and b, this can be rewritten in other forms, e.g.

$$\begin{aligned}\operatorname{sech}(x) - \operatorname{sech}(2x) &= \frac{\operatorname{sech}(x)(1-\operatorname{sech}(x))(2+\operatorname{sech}(x))}{2-\operatorname{sech}(x)^2}\\ \operatorname{sech}(x) - \operatorname{sech}(3x) &= \frac{4\operatorname{sech}(x)\tanh(x)^2}{4-3\operatorname{sech}(x)^2}\end{aligned}$$

Example. $f(x) = \arctan(\sinh(x))$

$$\int_0^\infty \frac{\arctan(\sinh(bx)) - \arctan(\sinh(ax))}{x}\,dx = \frac{\pi}{2}\log\left(\frac{b}{a}\right)$$

$$\begin{aligned}\arctan(\sinh(2x)) - \arctan(\sinh(x)) &= \arctan\left(\frac{\sinh(2x)-\sinh(x)}{1+\sinh(x)\sinh(2x)}\right)\\ &= \arctan\left(\frac{2\sinh(x)\cosh(x)-\sinh(x)}{1+\sinh(x)(2\sinh(x)\cosh(x))}\right)\\ &= \arctan\left(\frac{\sinh(x)(2\cosh(x)-1)}{1+2\sinh(x)^2\cosh(x)}\right)\end{aligned}$$

The last part of the proof of Frullani's theorem can be adapted to obtain a more general result:

Theorem 0.2. Suppose $f \in C[0,\infty)$, $\lim_{x\to\infty}\frac{1}{x}\int_0^x f(y)\,dy = L$ and $a, b > 0$. Then:

$$\int_0^\infty \frac{f(bx) - f(ax)}{x}\,dx = (L - f(0))\log\left(\frac{b}{a}\right)$$

Example. $f(x) = \cos(x)$

$$\int_0^\infty \frac{\cos(bx) - \cos(ax)}{x}\,dx = -\log\left(\frac{b}{a}\right)$$

In this case, $\lim_{x\to\infty}\frac{1}{x}\int_0^x \cos(y)\,dy = \lim_{x\to\infty}\frac{\sin(x)}{x} = 0$ while $\lim_{x\to\infty}\cos(x)$ does not exist.

Another way to evaluate this integral is to use Frullani's theorem with $f(x) = e^{-px}\cos(x)$ for $p > 0$, then take the limit as $p \to 0^+$. However, this requires care as the integral is not absolutely convergent for $p = 0$.

Example. Suppose $p, q \in \mathbb{R}$, $p > |q|$. Then we have:

$$\int_0^\infty \frac{\sin(px)\sin(qx)}{x}\,dx = \frac{1}{2}\int_0^\infty \frac{\cos((p-q)x) - \cos((p+q)x)}{x}\,dx = \frac{1}{2}\log\left(\frac{p+q}{p-q}\right) = \operatorname{arctanh}\left(\frac{q}{p}\right)$$

$$\int_0^\infty \frac{\operatorname{sech}(x)(1-\operatorname{sech}(x))(2+\operatorname{sech}(x))}{x\left(2-\operatorname{sech}(x)^2\right)}\,dx = \log(2)$$

$$\int_0^\infty \frac{1}{x}\arctan\left(\frac{\sinh(x)(2\cosh(x)-1)}{1+2\sinh(x)^2\cosh(x)}\right)dx = \frac{\pi}{2}\log(2)$$

$$\int_0^\infty \frac{1}{x}\arccos\left(\frac{\cosh(4x)-\cosh(2x)+2}{2\cosh(x)\cosh(3x)}\right)dx = \frac{\pi}{2}\log(3)$$

$$\int_0^1 \frac{x}{(1+3x^2)\log\left(\frac{1+x}{1-x}\right)}\,dx = \frac{\log(3)}{4}$$

More examples are available at `https://scientia.mat.utfsm.cl/archivos/vol19/vol19art14.pdf`.

Theorem 0.3. Suppose $f \in C[0,\infty)$, $\lim_{x\to\infty}\frac{1}{x}\int_0^x f(y)\,dy = L$ and $a, b > 0$. Then:

$$\int_0^\infty \frac{f(bx)-f(ax)}{x}\,dx = (L - f(0))\log\left(\frac{b}{a}\right)$$

Sub $u = e^x$ and $g(u) = f(\log(u))$, $\frac{1}{x}\int_1^{e^x} g(v)/v\,dv = L$

$$\int_1^\infty \frac{g(u^b)-g(u^a)}{u\log(u)}\,du = (L - g(1))\log\left(\frac{b}{a}\right)$$

Sub $u = e^{-x}$ and $g(u) = f(-\log(u))$, $-\frac{1}{x}\int_1^{e^{-x}} f(v)/v\,dv = L$

$$\int_0^1 \frac{g(u^b)-g(u^a)}{u\log(u)}\,du = (L - g(1))\log\left(\frac{b}{a}\right)$$

Corollary 0.4. Suppose $f \in C[1,\infty)$, $a, b > 0$ and $\lim_{x\to\infty}\frac{1}{\log(x)}\int_1^x \frac{f(y)}{y}\,dy = L$ exists and is finite. Then:

$$\int_1^\infty \frac{f(x^b)-f(x^a)}{x\log(x)}\,dx = (L - f(1))\log\left(\frac{b}{a}\right)$$

Proof. Define $g : [0,\infty) \to \mathbb{R}$ by $g(u) = f(e^u)$. Then $g \in C[0,\infty)$. Substituting $u = \log(x)$ into the integral yields:

$$\int_1^\infty \frac{f(x^b)-f(x^a)}{x\log(x)}\,dx = \int_0^\infty \frac{f\left(e^{bu}\right)-f\left(e^{au}\right)}{u}\,du = \int_0^\infty \frac{g(bu)-g(au)}{u}\,du$$

Similarly, substituting $u = \log(x)$ and $v = \log(y)$ into the limit condition yields:

$$\frac{1}{\log(x)}\int_1^x \frac{f(y)}{y}\,dy = \frac{1}{u}\int_1^{e^u}\frac{f(y)}{y}\,dy = \frac{1}{u}\int_0^u f(e^v)\,dv = \frac{1}{u}\int_0^u g(v)\,dv$$

Thus $\lim_{u\to\infty}\frac{1}{u}\int_0^u g(v)\,dv = \lim_{x\to\infty}\frac{1}{\log(x)}\int_1^x \frac{f(y)}{y}\,dy = L$. Applying **Theorem 0.2** to g yields:

$$r$$

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Suppose $f(x) = \sin(\log(x))$.

$$\int_1^x \sin$$

Suppose $\lim_{x\to\infty}\frac{1}{x}\int_0^x f(y)\,dy = L$. Substituting

$$\frac{1}{x}\int_0^x f(y)\,dy$$

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Proposition A.1.